

A Four-Sector Resummation Theorem for Recursive Green-Function Problems

Propagation, Interaction, Holonomy, and Recursive Closure from Dyson Equations to Aharonov–Bohm Networks

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Abstract

Many apparently different problems in theoretical physics share a common structural pattern: propagation between states, local interaction or scattering, phase accumulation around closed cycles, and the repeated generation of admissible histories. This work introduces a four-component architectural framework based on propagation, interaction, holonomy or closed-path monodromy, and recursive closure. The first three components define the primitive internal dynamics, while recursive closure organizes arbitrarily many repeated processes into a finite operator representation. A general resummation theorem is established for systems in which admissible histories can be uniquely factorized into primitive return blocks. Under this condition, the infinite family of repeated histories is represented exactly by a finite resolvent acting on an internal architectural state space. The resulting framework is then applied to several well-known physical problems, including Dyson resummation, Lippmann–Schwinger scattering, the Bethe–Salpeter equation, Foldy–Lax multiple scattering, Fabry–Pérot interference, quantum double-barrier resonant tunneling, and Aharonov–Bohm ring transport. These examples show that physically distinct systems can share the same recursive architecture despite differences in their microscopic interpretation. In particular, Fabry–Pérot interference and quantum double-barrier tunneling are shown to possess the same scalar recursive structure, while the Aharonov–Bohm ring provides a nontrivial realization in which closed-path information carries genuine gauge holonomy. The individual physical solutions and the underlying resolvent identities are established results. The contribution of the present framework is their systematic sector-resolved factorization into a common architectural calculus and the associated interpretation of recursive closure as an exact compression mechanism for infinite families of physical histories. The approach does not claim a fundamental reconstruction of quantum mechanics, but proposes a unified mathematical language for a broad class of recursive Green-function, scattering, interference, and network problems.

Keywords: Green functions; resummation; Dyson equation; Lippmann–Schwinger equation; Bethe–Salpeter equation; multiple scattering; Fabry–Pérot interferometer; resonant tunneling; Aharonov–Bohm effect; quantum graphs; holonomy; resolvent

1. Introduction

Recursive expansions occur throughout mathematical and theoretical physics. Dyson’s formulation of quantum electrodynamics provided one of the best-known examples in which repeated interaction insertions reorganize perturbative contributions into a dressed propagator [1]. Schwinger’s Green-function formulation established Green functions as fundamental tools for representing quantized fields and their dynamics [2]. Feynman’s space-time formulation

simultaneously emphasized the complementary idea that physical amplitudes may be organized as sums over histories or paths [3].

In scattering theory, the Lippmann–Schwinger construction expresses the full scattering state or transition operator through repeated free propagation and interaction [4]. Closely related recursive structures appear in multiple-wave scattering, beginning with Foldy’s treatment of isotropic scatterers [5], Lax’s general multiple-scattering formulation [6], its self-consistent extension to dense systems [7], and Watson’s many-body multiple-scattering expansion [8].

Relativistic two-particle physics produces a formally analogous structure. The Bethe–Salpeter equation generates repeated propagation of a particle pair between irreducible interaction kernels [9]. Although the physical interpretation differs substantially from ordinary potential scattering, the algebraic recurrence has a recognizable common skeleton.

Recursive histories also appear far outside quantum field theory. Fabry and Perot’s interferometric construction generates an infinite sequence of multiply reflected optical beams [10]. In semiconductor physics, tunneling through finite superlattices was analyzed by Tsu and Esaki [11], and resonant transmission through semiconductor double barriers was subsequently formulated explicitly by Chang, Esaki, and Tsu [12]. The multiple-reflection structure is strikingly analogous to an optical Fabry–Pérot cavity.

Closed-path phases introduce another ingredient. The Aharonov–Bohm effect demonstrates that a gauge potential can produce an observable phase associated with a closed path even when the particle propagates through a field-free region [13]. Byers and Yang established the associated flux-periodicity structure for multiply connected systems [14]. Mesoscopic $\hbar/e$ Aharonov–Bohm oscillations were subsequently observed in normal-metal rings [15], while multichannel transport through such rings was formulated within scattering theory by Büttiker, Imry, Landauer, and Pinhas [16].

The relation between Green functions and measurable transmission was developed further through the Fisher–Lee relation [17], building on the transmission-based transport viewpoint introduced by Landauer [18]. Network formulations therefore connect Green functions, scattering matrices, propagation amplitudes, and experimentally accessible transport observables.

Quantum graphs provide an especially transparent mathematical setting in which these elements coexist. Kottos and Smilansky showed that networks can be quantized in a manner that exposes their path and spectral structure [19], and subsequently developed an exact periodic-orbit formulation of quantum-graph spectra [20]. Kostykin and Schrader formulated general scattering theory and composition rules for quantum wires represented as graphs [21], while Barra and Gaspard studied transport and dynamics in open quantum graphs [22].

Recursive compression is also familiar from numerical Green-function methods. Haydock, Heine, and Kelly developed an influential recursion method for electronic-structure calculations [23]. For quantum graphs, exact Green functions can be organized as sums over scattering paths and reduced to closed analytic expressions [24]. An adjacency-matrix construction was later shown to be unitarily equivalent to the Green-function formulation [25]. Exact convergent periodic-orbit expansions have been obtained for regular quantum graphs [26], and analytic summations over families of graph trajectories have been used to describe Anderson localization [27]. Open graphs additionally provide exact scattering matrices and resonance structures in terms of graph dynamics [28].

The role of the closed-path sector becomes still clearer from gauge theory. Wu and Yang formulated gauge fields in terms of nonintegrable phase factors associated with paths [29]. Berry demonstrated that cyclic evolution can generate an additional geometric phase [30], while Wilczek and Zee showed that cyclic evolution can support non-Abelian gauge structure [31].

The existence of these many mathematically related constructions raises a structural question:

Can propagation, local interaction, closed-path phase information, and repeated-history summation be separated into a common minimal calculus that applies across these otherwise different physical problems?

The present paper investigates this question.

We introduce four architectural components:

P = propagation,

I = interaction or local scattering,

H = closed-path monodromy or holonomy,

and

R = recursive closure.

A central refinement is that these four objects are **not treated symmetrically**. The first three generate elementary return processes. Recursive closure then acts on the resulting kernel:

$$\boxed{(P, I, H) \rightarrow K_{PIH} \rightarrow R(K_{PIH}).}$$

Thus the fundamental operational structure is more accurately written

$$\boxed{\mathcal{A} = R(P, I, H)}$$

than as a simple four-term direct sum.

We prove a general resummation theorem for systems whose admissible histories possess a unique decomposition into primitive return blocks. We then show that several famous equations arise as realizations of this single structure.

The purpose is **not** to claim that the individual solutions are new. They are not. Nor is the Neumann-series identity new.

The proposed contribution is instead a sector-resolved structural statement:

$$\boxed{\text{many apparently different recursive physical constructions} \rightarrow \text{one architectural factorization.}}$$

The strongest consequence of this viewpoint is a compression principle:

$$\boxed{\text{infinitely many histories} \rightarrow \text{finite internal kernel} \rightarrow \text{one resolvent.}}$$

2. The Four Architectural Components

2.1 Propagation sector P

Let $\mathcal{X}$ denote an internal state space.

A propagation operator

$$P(z)$$

transports amplitudes between distinguished interaction, scattering, or junction states.

The parameter z may represent energy, momentum, frequency, or another spectral variable.

Examples include

$$P(E) = G_0(E)$$

for a free Green function,

$$P = e^{ikL}$$

for free propagation through a one-dimensional interval, and

$$P_{ij} = G_0(\mathbf{r}_i, \mathbf{r}_j)$$

for propagation between spatially separated scatterers.

Propagation need not be scalar and need not commute with the other architectural sectors.

2.2 Interaction sector I

The interaction sector contains irreducible local modifications of propagation.

Examples include

$$I = \Sigma$$

for a self-energy insertion,

$$I = V$$

for a potential,

$$I = K$$

for a Bethe–Salpeter irreducible kernel, and local reflection, transmission, or vertex-scattering matrices in network problems.

The adjective *irreducible* is important.

If a process already contains arbitrary recursive repetition, it belongs conceptually to the closed architecture rather than to the primitive interaction sector.

2.3 Holonomy sector H

The third sector carries information associated with closed propagation cycles.

In its most general form, let

$$H: \mathcal{C} \rightarrow GL(\mathcal{X})$$

assign a monodromy operator to a closed cycle C .

For an Abelian gauge phase,

$$H(C) = \exp \left[\frac{iq}{\hbar} \oint_C \mathbf{A} \cdot d\mathbf{l} \right].$$

For a simple optical cavity, H may instead represent a closed-cycle dynamical monodromy such as

$$H = e^{2ikL}.$$

These two examples should not be conflated.

The first is genuine gauge holonomy.

The second is ordinary closed-orbit phase accumulation.

We therefore use **closed-path monodromy** as the generic mathematical term and reserve **gauge holonomy** for the corresponding gauge-theoretical realization.

For systems without a relevant closed-cycle phase,

$$H = \mathbb{1}.$$

2.4 Recursive sector R

The fourth sector is qualitatively different.

It is not another propagation path or another local interaction.

It is an operation acting on a return kernel.

For an operator K , define

$$R(K) = \sum_{n=0}^{\infty} K^n$$

whenever this series converges.

If

$$\|K\| < 1,$$

then

$$\boxed{R(K) = (\mathbb{1} - K)^{-1}.$$

Thus R is a **closure operation**.

Its role is to replace explicit enumeration of arbitrarily many repeated histories by a resolvent.

3. Primitive Return Blocks

Suppose an observable process begins in an entrance sector, enters an internal region, may execute zero or more internal return processes, and finally exits.

Let

$$E: \mathcal{X}_{\text{in}} \rightarrow \mathcal{X}$$

be the entrance map and

$$F: \mathcal{X} \rightarrow \mathcal{X}_{\text{out}}$$

the exit map.

Let

$$\mathfrak{B} = \{\beta_1, \dots, \beta_m\}$$

be a finite set of **primitive return blocks**.

Each primitive block begins and ends in the internal state space $\mathcal{X}$.

Its amplitude is

$$B_\alpha = A(\beta_\alpha) \in \mathcal{B}(\mathcal{X}).$$

Every B_α is constructed from the algebra generated by the three primitive sectors:

$$B_\alpha \in \text{Alg}(P, I, H).$$

Define the total primitive return kernel

$$K_{PIH} = \sum_{\alpha=1}^m B_\alpha.$$

In a one-channel problem there may be only one return block, in which case

$$K_{PIH} = B_1.$$

Frequently this takes the schematic form

$$K_{PIH} = IPH,$$

but this ordering is **not universal**.

For noncommuting operators the ordering is determined by the physical sequence of events.

4. Unique Factorization of Histories

The crucial assumption is not merely the existence of a geometric series.

It is the existence of a controlled decomposition of histories.

Definition 1 — PIH-factorizable history system

A history system is called **PIH-factorizable** if every non-direct admissible history from input to output can be written uniquely as

$$\gamma = \gamma_{\text{out}} \circ \beta_{\alpha_n} \circ \cdots \circ \beta_{\alpha_2} \circ \beta_{\alpha_1} \circ \gamma_{\text{in}},$$

where

$$\beta_{\alpha_j} \in \mathfrak{B}.$$

The integer

$$n \geq 0$$

is the recursive depth.

Uniqueness means that no physical history is counted more than once.

This requirement is important because an arbitrary grouping of paths could otherwise produce overcounting.

In graph problems, uniqueness can usually be restored by enlarging the internal state space from vertices to **directed bonds** or other suitably refined states.

5. PIHR Resummation Theorem

Theorem 1 — PIHR Resummation Theorem

Let $\mathcal{X}$ be a Banach space and let a physical history system be PIH-factorizable with bounded entrance map E , exit map F , and finite primitive return family

$$\mathfrak{B} = \{\beta_1, \dots, \beta_m\}.$$

Let

$$K_{PIH} = \sum_{\alpha=1}^m B_{\alpha}$$

be the associated return kernel.

Assume amplitudes are multiplicative under concatenation and

$$\|K_{PIH}\| < 1.$$

Then the complete contribution of all recursively generated histories is

$$\boxed{\mathcal{O}_{rec} = F(\mathbb{1} - K_{PIH})^{-1}E.}$$

If an additional direct contribution $\mathcal{O}_0$ exists outside the recursive internal region, then

$$\boxed{\mathcal{O} = \mathcal{O}_0 + F(\mathbb{1} - K_{PIH})^{-1}E.}$$

Proof

Histories containing exactly n primitive return blocks have amplitudes

$$F B_{\alpha_n} B_{\alpha_{n-1}} \cdots B_{\alpha_1} E.$$

Summing over all choices

$$\alpha_1, \dots, \alpha_n$$

gives

$$\sum_{\alpha_n, \dots, \alpha_1} F B_{\alpha_n} \cdots B_{\alpha_1} E.$$

By distributivity,

$$\sum_{\alpha_n, \dots, \alpha_1} B_{\alpha_n} \cdots B_{\alpha_1} = \left(\sum_{\alpha} B_{\alpha} \right)^n.$$

Therefore

$$\mathcal{O}^{(n)} = F K_{PIH}^n E.$$

Summing over all recursive depths,

$$\mathcal{O}_{rec} = \sum_{n=0}^{\infty} F K_{PIH}^n E = F \left(\sum_{n=0}^{\infty} K_{PIH}^n \right) E.$$

Because

$$\| K_{PIH} \| < 1,$$

the Neumann series converges in operator norm and

$$\sum_{n=0}^{\infty} K_{PIH}^n = (\mathbb{1} - K_{PIH})^{-1}.$$

Hence

$$\mathcal{O}_{rec} = F(\mathbb{1} - K_{PIH})^{-1}E.$$

■

6. Finite-Kernel Compression

The theorem immediately gives the main structural consequence.

Corollary 1 — Finite-Kernel Compression

If

$$\dim \mathcal{X} = d < \infty,$$

then an infinite family of recursively generated histories is represented exactly by the inverse of a $d \times d$ matrix:

$$\boxed{\{\text{infinitely many histories}\} \rightarrow K_{PIH} \in M_d(\mathbb{C}) \rightarrow (\mathbb{1} - K_{PIH})^{-1}.}$$

The number of histories may grow without bound while the architectural state dimension remains fixed.

This is the sense in which PIHR functions as a **compression calculus**.

The compression is exact, not asymptotic, provided the factorization assumptions hold.

7. Pole and Resonance Condition

Suppose $K_{PIH}(z)$ depends analytically on a complex spectral parameter z and acts on a finite-dimensional internal space.

Then

$$R(z) = (\mathbb{1} - K_{PIH}(z))^{-1}$$

is meromorphic.

Its singularities satisfy

$$\boxed{\det[\mathbb{1} - K_{PIH}(z)] = 0.}$$

Thus the same architectural kernel that performs resummation also determines resonances, bound-state conditions, or internal modes.

This observation will recur in every example below.

8. Architectural Equivalence

Definition 2 — Recursive architectural equivalence

Two recursively generated systems are said to be PIHR-equivalent if their internal kernels satisfy

$$K_2 = SK_1S^{-1}$$

for an invertible transformation S , with correspondingly transformed entrance and exit maps.

Corollary 2

If

$$K_2 = SK_1S^{-1},$$

then

$$R(K_2) = SR(K_1)S^{-1}.$$

Proof

$$\mathbb{1} - K_2 = S(\mathbb{1} - K_1)S^{-1}.$$

Taking inverses gives

$$(\mathbb{1} - K_2)^{-1} = S(\mathbb{1} - K_1)^{-1}S^{-1}.$$

■

Consequently, PIHR-equivalent systems possess the same internal pole condition.

This gives precise meaning to the statement that physically different systems can possess the same **recursive architecture**.

9. Example I: Dyson Resummation

The Dyson equation for a dressed propagator can be written schematically as

$$G = G_0 + G_0\Sigma G.$$

Iteration gives

$$G = G_0 + G_0\Sigma G_0 + G_0\Sigma G_0\Sigma G_0 + \cdots.$$

Define

$$P = G_0,$$

$$I = \Sigma,$$

and

$$H = \mathbb{1}.$$

The primitive return kernel is

$$K_D = \Sigma G_0 = IP.$$

Therefore

$$G = P[\mathbb{1} + K_D + K_D^2 + \cdots].$$

Applying R ,

$$\boxed{G = P R(IP)}$$

and hence

$$\boxed{G = G_0(\mathbb{1} - \Sigma G_0)^{-1}}.$$

Multiplication by G_0^{-1} gives the familiar equivalent form

$$\boxed{G^{-1} = G_0^{-1} - \Sigma}.$$

Thus the Dyson construction is the PIHR architecture

$$\boxed{(P, I, \mathbb{1}, R)}.$$

The holonomy sector is trivial.

The architecture does not replace Dyson's result; it identifies the recurrent structure as one realization of a broader return-kernel calculus.

For continuum quantum field theories the perturbative expansion may be formal and additional renormalization and domain questions arise. The theorem should therefore be interpreted literally only after an appropriate regularization or in an operator domain for which the required inverse exists.

10. Example II: Lippmann–Schwinger Equation

For a potential V , the transition operator satisfies

$$T = V + V G_0^{(+)} T.$$

Iterating,

$$T = V + V G_0^{(+)} V + V G_0^{(+)} V G_0^{(+)} V + \cdots.$$

Now identify

$$P = G_0^{(+)},$$

$$I = V,$$

$$H = \mathbb{1}.$$

Then

$$K_{LS} = PI = G_0^{(+)} V.$$

Therefore

$$T = I[\mathbb{1} + PI + (PI)^2 + \cdots],$$

so that

$$\boxed{T = I R(PI)}$$

or

$$\boxed{T = V(\mathbb{1} - G_0^{(+)} V)^{-1}}.$$

Using the push-through identity,

$$A(\mathbb{1} - BA)^{-1} = (\mathbb{1} - AB)^{-1}A,$$

one may equivalently write

$$T = (\mathbb{1} - VG_0^{(+)})^{-1}V.$$

Comparing with Dyson,

$$\text{Dyson: } G = P R(IP),$$

whereas

$$\text{Lippmann-Schwinger: } T = I R(PI).$$

The physical content is different, but the recursive architecture is the same.

11. Example III: Bethe–Salpeter Resummation

Let

$$\mathcal{P}_2 = G_1 G_2$$

denote the two-particle propagation operator and let $\mathcal{K}$ denote a two-particle irreducible interaction kernel.

The inhomogeneous Bethe–Salpeter scattering equation has the schematic form

$$\mathcal{T} = \mathcal{K} + \mathcal{K}\mathcal{P}_2\mathcal{T}.$$

Iteration yields

$$\mathcal{T} = \mathcal{K} + \mathcal{K}\mathcal{P}_2\mathcal{K} + \mathcal{K}\mathcal{P}_2\mathcal{K}\mathcal{P}_2\mathcal{K} + \cdots.$$

Identify

$$P = \mathcal{P}_2,$$

$$I = \mathcal{K},$$

$$H = \mathbb{1}.$$

Then

$$K_{BS} = PI = \mathcal{P}_2\mathcal{K}.$$

Therefore

$$\boxed{\mathcal{T} = I R(PI)}$$

and

$$\boxed{\mathcal{T} = \mathcal{K}(\mathbb{1} - \mathcal{P}_2\mathcal{K})^{-1}.}$$

Bound states appear when the resolvent becomes singular.

Thus the architectural pole condition is

$$\boxed{\det(\mathbb{1} - \mathcal{P}_2\mathcal{K}) = 0}$$

in a finite-dimensional representation, or the corresponding operator/Fredholm condition when mathematically defined.

The same architecture that generates ordinary scattering therefore also generates a bound-state condition through its recursive poles.

Again, PIHR does not solve the difficult problem of determining the irreducible Bethe–Salpeter kernel.

It separates that problem from the universal algebra of repeated propagation.

12. Example IV: Foldy–Lax Multiple Scattering

Consider N localized scatterers.

Let u_i be the exciting field at scatterer i , and $u_i^{(0)}$ the incident field.

A general multiple-scattering system has the structure

$$u_i = u_i^{(0)} + \sum_{j \neq i} G_{ij} t_j u_j,$$

where

$$G_{ij} = G_0(\mathbf{r}_i, \mathbf{r}_j)$$

propagates from scatterer j to scatterer i , and t_j is the local scattering amplitude.

Define the $N \times N$ propagation matrix

$$P_{ij} = (1 - \delta_{ij})G_{ij}$$

and interaction matrix

$$I = \text{diag}(t_1, \dots, t_N).$$

Then

$$\mathbf{u} = \mathbf{u}^{(0)} + PI \mathbf{u}.$$

Therefore

$$(\mathbb{1} - PI)\mathbf{u} = \mathbf{u}^{(0)},$$

and

$$\boxed{\mathbf{u} = R(PI)\mathbf{u}^{(0)} = (\mathbb{1} - PI)^{-1}\mathbf{u}^{(0)}}.$$

Expanding,

$$\mathbf{u} = \mathbf{u}^{(0)} + PI\mathbf{u}^{(0)} + (PI)^2\mathbf{u}^{(0)} + \dots.$$

The term

$$(PI)^n$$

contains all admissible sequences involving n additional inter-scatterer propagation/scattering steps.

This example makes the compression mechanism particularly transparent:

$$\boxed{\text{infinitely many multiple-scattering paths} \rightarrow N \times N \text{ kernel} \rightarrow \text{one matrix inverse.}}$$

The physical complexity has not disappeared: it is encoded in P and I .

But explicit enumeration of every multiple-scattering history has disappeared.

For ordinary scalar scattering,

$$H = \mathbb{1}.$$

If propagation around closed geometric cycles acquires additional phase or gauge information, H can be inserted as a separate monodromy sector.

13. Example V: Fabry–Pérot Interference

Consider two partially reflecting mirrors separated by a distance L .

Let

$$t_1, \quad t_2$$

be transmission amplitudes and

$$r_1', \quad r_2$$

the internal reflection amplitudes.

Single-pass propagation is

$$P = e^{ikL}.$$

The directly transmitted amplitude is

$$A_0 = t_1 P t_2.$$

A beam undergoing one additional round trip contributes

$$A_1 = t_1 P (r_1' r_2 P^2) t_2.$$

Two additional round trips give

$$A_2 = t_1 P (r_1' r_2 P^2)^2 t_2.$$

Thus

$$A_{FP} = t_1 P \sum_{n=0}^{\infty} (r_1' r_2 P^2)^n t_2.$$

The PIHR sectors are naturally

$$P = e^{ikL},$$

$$I_{FP} = r_1' r_2,$$

and

$$H_{FP} = e^{2ikL}.$$

Here H_{FP} is a **closed-cycle dynamical monodromy**, not a gauge holonomy.

The primitive return kernel is

$$K_{FP} = I_{FP} H_{FP} = r_1' r_2 e^{2ikL}.$$

Therefore

$$\boxed{A_{FP} = t_1 P R(K_{FP}) t_2}$$

and

$$A_{FP} = \frac{t_1 t_2 e^{ikL}}{1 - r_1' r_2 e^{2ikL}}.$$

For identical lossless mirrors with intensity reflectivity

$$\mathcal{R} = |r|^2$$

and round-trip phase δ , this leads to the standard Airy form

$$\mathcal{T}_{FP} = \frac{1}{1 + \frac{4\mathcal{R}}{(1 - \mathcal{R})^2} \sin^2(\delta/2)}.$$

Thus the famous Fabry–Pérot resonance is an elementary scalar PIHR closure.

14. Example VI: Quantum Double-Barrier Resonant Tunneling

Now replace the two optical mirrors by two quantum-mechanical potential barriers.

Let

$$t_1, \quad t_2$$

be their transmission amplitudes and

$$r_1', \quad r_2$$

the reflection amplitudes seen from the intermediate region.

Between the barriers the wave propagates through a distance L :

$$P = e^{ikL}.$$

The direct transmission amplitude is

$$t^{(0)} = t_1 P t_2.$$

After one internal round trip,

$$t^{(1)} = t_1 P (r_1' r_2 P^2) t_2.$$

After n round trips,

$$t^{(n)} = t_1 P (r_1' r_2 P^2)^n t_2.$$

Hence

$$t_{DB} = t_1 P \sum_{n=0}^{\infty} (r_1' r_2 P^2)^n t_2.$$

Therefore

$$t_{DB} = t_1 P R (r_1' r_2 P^2) t_2$$

and

$$t_{DB} = \frac{t_1 t_2 e^{ikL}}{1 - r_1' r_2 e^{2ikL}}.$$

The architectural identity with the Fabry–Pérot cavity is immediate:

$$\boxed{K_{DB} = K_{FP}}$$

after replacing optical mirror amplitudes by quantum-barrier amplitudes.

Consequently,

$$\boxed{\text{Fabry–Pérot optical resonance} \quad \text{and} \quad \text{quantum double-barrier resonance}}$$

belong to the same scalar recursive architecture.

Their resonance condition is

$$1 - r_1' r_2 e^{2ikL} = 0$$

in the idealized pole problem, or, for real scattering energies, maximal transmission occurs when the round-trip phase satisfies the appropriate constructive-interference condition.

This is a particularly transparent example of cross-domain architectural equivalence.

The underlying physical theories remain different.

The recursive skeleton is the same.

15. Example VII: Aharonov–Bohm Ring

The previous examples either have

$$H = \mathbb{1}$$

or a dynamical round-trip monodromy.

An Aharonov–Bohm ring provides a case in which H carries genuine gauge information.

Consider two junctions, left L and right R , connected by upper and lower arms.

The internal state space is

$$\mathcal{X} = \mathbb{C}^2,$$

corresponding to amplitudes on the two directed arms.

Define forward dynamical propagation

$$P_+ = \begin{pmatrix} e^{ikL_u} & 0 \\ 0 & e^{ikL_d} \end{pmatrix}$$

and backward propagation

$$P_- = \begin{pmatrix} e^{ikL_u} & 0 \\ 0 & e^{ikL_d} \end{pmatrix}.$$

Introduce gauge-phase matrices

$$H_+ = \begin{pmatrix} e^{i\chi_u} & 0 \\ 0 & e^{i\chi_d} \end{pmatrix},$$

$$H_- = H_+^\dagger.$$

The gauge-invariant closed-loop phase is

$$\chi_u - \chi_d = \varphi_{AB},$$

with

$$\varphi_{AB} = \frac{q}{\hbar} \oint \mathbf{A} \cdot d\mathbf{l} = 2\pi \frac{\Phi}{\Phi_0}.$$

Let

$$J_L, \quad J_R$$

be the 2×2 internal reflection/mixing submatrices of the two junctions.

Starting immediately after the left junction, one complete internal return cycle consists of

1. forward propagation to the right junction,
2. scattering at the right junction,
3. backward propagation,
4. scattering at the left junction.

The return kernel is therefore

$$K_{AB} = J_L P_- H_- J_R P_+ H_+.$$

Let

$$a \in \mathbb{C}^2$$

be the injection vector from the incoming lead into the two arms, and let

$$b^\dagger$$

extract the outgoing amplitude at the right junction.

Transmission with no additional round trip is

$$t_0 = b^\dagger P_+ H_+ a.$$

One additional internal round trip contributes

$$t_1 = b^\dagger P_+ H_+ K_{AB} a.$$

After n round trips,

$$t_n = b^\dagger P_+ H_+ K_{AB}^n a.$$

Summing all paths,

$$t_{AB} = b^\dagger P_+ H_+ \sum_{n=0}^{\infty} K_{AB}^n a.$$

Hence

$$t_{AB} = b^\dagger P_+ H_+ R(K_{AB}) a$$

or

$$t_{AB} = b^\dagger P_+ H_+ (\mathbb{1} - K_{AB})^{-1} a.$$

Here every PIHR component is essential:

P :dynamical propagation along the arms,

I :junction scattering J_L, J_R ,

H :Aharonov–Bohm gauge phase,

R :sum over arbitrarily many windings.

The resonances satisfy

$$\boxed{\det(\mathbb{1} - K_{AB}) = 0.}$$

This is the cleanest nontrivial realization of the complete four-component architecture.

16. General Quantum-Graph Formulation

The Aharonov–Bohm ring generalizes naturally.

Let a finite open quantum graph possess B internal bonds.

It is often useful to work in the $2B$ -dimensional directed-bond space.

Let

$$S$$

denote the vertex-scattering operator.

Let

$$P(k)$$

be the diagonal propagation matrix containing factors

$$e^{ikL_b}.$$

Let

$$H(\Phi)$$

contain the directed magnetic or other closed-path phases.

Define the internal one-step map

$$\boxed{U(k, \Phi) = S P(k) H(\Phi)}$$

up to the convention chosen for operator ordering.

Then a path involving n internal steps contributes a factor

$$U^n.$$

The complete internal history sum is therefore

$$\boxed{R(U) = (\mathbb{1} - U)^{-1}.}$$

For an open graph,

$$\mathcal{S}_{eff} = \mathcal{S}_0 + W_{out}(\mathbb{1} - U)^{-1}W_{in},$$

where W_{in} and W_{out} connect external leads to the internal directed-bond space.

Thus an exponentially proliferating or infinite path family is compressed to a $2B \times 2B$ matrix inverse.

For a closed graph the spectral condition takes the familiar form

$$\det[\mathbb{1} - U(k, \Phi)] = 0.$$

This is precisely the type of example for which the architectural interpretation has the greatest potential usefulness:

$$\text{path proliferation} \rightarrow \text{finite internal state space.}$$

17. A Common Master Form

The examples can now be placed into one general expression.

Let

$$K_{PIH}$$

be the primitive internal return kernel.

Then an observable has the architecture

$$\mathcal{O} = \mathcal{O}_0 + F R(K_{PIH}) E$$

with

$$R(K_{PIH}) = (\mathbb{1} - K_{PIH})^{-1}.$$

The important object is therefore not the geometric series itself.

It is the factorization

$$\text{physical histories} \rightarrow (P, I, H) \rightarrow K_{PIH} \rightarrow R(K_{PIH}).$$

The same form occurs in systems with radically different physical interpretations.

18. Cross-Domain Comparison

Problem	Propagat ion P	Interaction I	Holonomy/m onodromy H	Recursive closure R	Result
Dyson	G_0	Σ	1	repeated self- energy insertions	$G = G_0 R(\Sigma G_0)$
Lippmann– Schwinger	$G_0^{(+)}$	V	1	repeated scattering	$T = V R G_0 + V$
Bethe– Salpeter	$G_1 G_2$	irreducible kernel K	1	repeated two- body propagation	$T = K R(P_2 K)$
Foldy–Lax	G_{ij}	t_i	usually 1	arbitrary multiple scattering	$u = R P I u^{(0)}$
Fabry– Pérot	$e^{(ikL)}$	mirror reflections	$e^{(2ikL)}$	arbitrary round trips	$t \propto R(r_1' r_2 e^{(2ikL)})$
Double barrier	$e^{(ikL)}$	barrier reflections	$e^{(2ikL)}$	arbitrary internal reflections	$t \propto R(r_1' r_2 e^{(2ikL)})$
Aharonov– Bohm ring	arm propagati on	junction scattering	$e^{(i\varphi_{AB})}$	arbitrary windings	$t \sim (1 - K_A B)^{-1}$

Problem	Propagation P	Interaction I	Holonomy/monodromy H	Recursive closure R	Result
General quantum graph	bond propagation	vertex matrix S-	cycle phases	arbitrary graph paths	$(1 - U)^{-1}$

Three structural classes are immediately visible.

First,

$$H = \mathbb{1}$$

for Dyson, Lippmann–Schwinger, and the elementary Bethe–Salpeter and Foldy–Lax cases.

Second, optical and double-barrier cavities possess closed-cycle dynamical monodromy.

Third, flux-threaded networks possess genuine gauge holonomy.

Thus the H sector is capable of distinguishing topology-sensitive recursions from otherwise similar local resummations.

19. Fabry–Pérot and Double-Barrier Tunneling as an Architectural Pair

The closest cross-domain correspondence is

Fabry–Pérot optics $\leftrightarrow$ double-barrier quantum tunneling.
--

For both,

$$K = r_1' r_2 e^{2ikL}.$$

For both,

$$R(K) = \frac{1}{1 - K}.$$

For both, resonance follows from phase closure of the internal round trip.

The distinction lies in the physical origin of the amplitudes:

- electromagnetic boundary conditions in the optical cavity;
- quantum-mechanical barrier matching in the tunneling problem.

From the architectural perspective these two physical systems lie in the same scalar recursive equivalence class.

This is an example of what the PIHR language is intended to expose:

different physics but the same recursive architecture.
--

20. Dyson, Lippmann–Schwinger, and Bethe–Salpeter as an Architectural Family

A second family consists of integral-equation resummations.

Dyson:

$$G = P R(IP).$$

Lippmann–Schwinger:

$$T = I R(PI).$$

Bethe–Salpeter:

$$\mathcal{T} = I R (PI).$$

The distinctions between the theories reside principally in:

1. the space on which P acts;
2. the physical meaning of I ;
3. the irreducibility condition imposed on I ;
4. the analytic and renormalization properties of the resulting operators.

The recursive algebra is shared.

This suggests viewing the resolvent as a **structural closure operator** rather than as a formula specific to any one of these theories.

21. Why R Is Different from P , I , and H

The examples reveal an important point about the original four-sector picture.

Propagation, interaction, and holonomy describe ingredients of a primitive history.

Recursive closure describes what is done with those ingredients.

Thus

$$P, I, H$$

are architectural data, whereas

$$R$$

is an architectural operation.

The hierarchy is therefore

$$\boxed{(P, I, H) \xrightarrow{\text{factorization}} K_{PIH} \xrightarrow{R} (\mathbb{1} - K_{PIH})^{-1}.}$$

This asymmetry is not a defect.

It makes the framework more precise.

22. Architecture as a Compression Principle

Suppose a network has internal state dimension

$$d.$$

At recursive depth n , the number of distinct admissible histories may grow roughly as

$$N_n \sim \mu^n$$

for some branching factor $\mu > 1$.

Explicit history enumeration through depth n therefore becomes exponentially expensive.

Yet the complete sum may still be represented by

$$(\mathbb{1} - K_{PIH})^{-1}$$

with

$$K_{PIH} \in M_d(\mathbb{C}).$$

The internal dimension remains

$$d$$

while

$$N_n \rightarrow \infty.$$

Hence the architectural reduction is conceptually

$$\boxed{\lim_{n \rightarrow \infty} N_n = \infty \quad \text{but} \quad \dim K_{PIH} = d < \infty.}$$

This is the strongest computational interpretation of the framework.

Importantly, the framework does **not** guarantee that finding the correct finite kernel is always easy.

The difficult part may be the construction of a nonredundant internal state space and the proof of unique factorization.

23. When the Method Fails

The PIHR reduction is not universal without qualifications.

It can fail for several reasons.

23.1 Nonunique decomposition

If one physical history can be decomposed into primitive blocks in several inequivalent ways, the naive expansion overcounts histories.

The internal state space must then be refined.

23.2 Infinite irreducible memory

Some systems have memory that cannot be represented by a finite internal state.

Then no finite-dimensional K_{PIH} exists.

23.3 Strong nonlocality

If the interaction sector depends on the entire previous path rather than on the current internal state, the Markov-like kernel representation may fail.

23.4 Divergent recursive expansion

The Neumann series does not converge when the relevant operator lies outside its convergence domain.

One may still obtain a resolvent by analytic continuation, but this requires additional mathematical justification.

23.5 Singular continuum operators

Quantum-field and continuum integral equations may involve unbounded operators, ultraviolet divergences, renormalization, or singular kernels.

The finite-dimensional theorem cannot simply be transferred without domain analysis.

24. What Is New and What Is Established

A mathematically responsible statement of novelty is essential.

The following are **not new**:

$$\sum_{n=0}^{\infty} K^n = (\mathbb{1} - K)^{-1},$$

the Dyson equation,

the Lippmann–Schwinger equation,

the Bethe–Salpeter equation,

Foldy–Lax multiple scattering,

Fabry–Pérot resummation,

double-barrier resonant tunneling,

Aharonov–Bohm interference,

or quantum-graph scattering theory.

The proposed new element is the specific **four-component architectural organization**

$$P, I, H, R$$

with the structural hierarchy

$$\boxed{(P, I, H) \rightarrow K_{PIH} \rightarrow R(K_{PIH}),}$$

together with the explicit proposition that this hierarchy constitutes a common reduction calculus across these domains.

The paper therefore makes a **structural unification claim**, not a claim to new solutions of the individual classical problems.

25. A Stronger Architectural Conjecture

The examples suggest a broader question.

Conjecture 1 — Finite Architectural Closure Conjecture

Let a linear physical theory possess:

1. a finite set of internal boundary states;
2. multiplicative amplitudes under path concatenation;
3. local scattering or interaction maps;
4. closed-path monodromy represented on the internal state space;
5. no irreducible dependence on the complete prior history.

Then its exact repeated-history contribution can be represented by a finite PIH kernel and recursive closure

$$\boxed{R(K_{PIH}) = (\mathbb{1} - K_{PIH})^{-1}.}$$

The conjectural content is not the matrix identity.

It is the claim that a broad physically relevant class admits a **finite PIH factorization**.

This is the point that requires future mathematical classification.

26. Minimal Architectural State Space

A particularly important open problem is whether the internal state dimension can be minimized canonically.

Let

$$d_{\min}$$

denote the smallest dimension of an internal space capable of representing all admissible recursive histories without ambiguity.

Then

$$d_{\min}$$

could itself become an architectural invariant.

Two apparently different systems could possess the same minimal kernel dimension and equivalent return operators.

This suggests a program in which architectures are classified not by individual paths but by minimal recursive representations.

27. Relation to the Earlier Four-Sector Architecture

The present result also modifies the interpretation of a four-sector architecture.

Rather than

$$\mathcal{A} = \mathcal{A}_P \oplus \mathcal{A}_I \oplus \mathcal{A}_H \oplus \mathcal{A}_R,$$

the examples favor

$$\boxed{\mathcal{A} = R(\mathcal{A}_P, \mathcal{A}_I, \mathcal{A}_H).}$$

This distinction is significant.

The geometric sectors

$$P, I, H$$

specify what an elementary history can do.

The recursive sector specifies how those histories close under repetition.

In that sense,

$$\boxed{R = \text{closure of architectural possibility.}}$$

28. Potential Extension to Non-Abelian Holonomy

Nothing in the finite-kernel theorem requires H to be a scalar phase.

For a non-Abelian holonomy,

$$H(C) \in U(N),$$

different loop operations need not commute.

The primitive kernel then becomes genuinely matrix ordered:

$$K_{PIH} = \sum_{\alpha} I_{\alpha_m} P_{\alpha_m} H_{\alpha_m} \cdots I_{\alpha_1} P_{\alpha_1} H_{\alpha_1}.$$

Recursive closure remains

$$R(K_{PIH}) = (\mathbb{1} - K_{PIH})^{-1},$$

but the spectrum and pole structure can depend on noncommuting loop data.

This extension may provide a more demanding test of the architecture than the Abelian examples considered here.

29. Potential Extension to Dyson–Schwinger Hierarchies

The elementary Dyson equation closes on a single dressed propagator.

Full Dyson–Schwinger systems are more complex because different n -point functions couple to one another.

A finite closed truncation could nevertheless be represented by an enlarged architectural state vector

$$\mathbf{G} = \begin{pmatrix} G_2 \\ G_3 \\ \vdots \\ G_N \end{pmatrix}.$$

If the recursive part is linearized or represented by a finite block kernel,

$$\mathbf{G} = \mathbf{G}_0 + K_{PIH} \mathbf{G},$$

then

$$\mathbf{G} = (\mathbb{1} - K_{PIH})^{-1} \mathbf{G}_0.$$

The nonlinear case is substantially harder and is not covered by Theorem 1.

Whether useful nonlinear generalizations of R exist is an open problem.

30. Physical Interpretation

The cross-domain examples suggest the following operational meanings:

P : Where can amplitude propagate?

I : What happens locally when it arrives?

H : What information is accumulated by closing a cycle?

R : What happens when the allowed cycle is repeated arbitrarily often?

This formulation is intentionally modest.

It does not assert that these four questions constitute all of fundamental physics.

It asserts that they organize a broad and important class of recursive linear problems.

31. Discussion

The principal result of this paper is the identification of a common architecture behind several historically and physically distinct constructions.

At first sight, Dyson dressing, quantum scattering, relativistic two-particle propagation, multiple-wave scattering, optical cavity interference, semiconductor resonant tunneling, and flux-threaded quantum networks belong to different branches of physics.

Their primitive objects are different.

Their observables are different.

Their physical interpretations are different.

Nevertheless, once the irreducible physical content has been isolated, each contains a repeated-history structure.

In the simplest cases this structure reduces to a scalar geometric series.

In network and multiple-scattering problems it becomes a finite-dimensional matrix resolvent.

The PIHR formulation separates four roles that are frequently mixed together:

- propagation determines connectivity and dynamical transfer;
- interaction determines local modification;
- holonomy determines closed-cycle information;
- recursion sums all admissible repeated histories.

The most important mathematical refinement is recognizing that recursion is not a fourth local building block.

Instead,

$$R$$

acts on a kernel assembled from

$$P, I, H.$$

The resulting hierarchy

$$(P, I, H) \rightarrow K_{PIH} \rightarrow R(K_{PIH})$$

is more precise than a symmetric four-sector decomposition.

The Foldy–Lax and quantum-graph examples illustrate why this viewpoint may be computationally useful. Explicit path families can proliferate indefinitely, while the internal kernel remains finite dimensional.

This yields the architecture-compression principle

$\text{history complexity} \neq \text{kernel dimension.}$

A system can therefore possess infinitely many histories without requiring an infinite explicit enumeration of them.

The Fabry–Pérot/double-barrier correspondence provides a second type of evidence. These systems belong to different physical domains but reduce to the identical scalar recursive kernel

$$r_1' r_2 e^{2ikL}.$$

The Aharonov–Bohm example adds a qualitatively new feature because the closed-cycle sector carries gauge information that cannot be reduced to local scattering amplitudes alone.

Hence

$$H$$

becomes genuinely independent in multiply connected gauge-sensitive systems.

32. Limitations

Several limitations should prevent overinterpretation.

First, the core resummation theorem is mathematically a structured application of the Neumann resolvent.

Its value, if any, lies in the **factorization principle**, not in a new matrix identity.

Second, many areas of physics already use transfer matrices, scattering matrices, recursion methods, graph reductions, and Green-function resolvents. PIHR must therefore demonstrate practical or conceptual simplification beyond a change of notation.

Third, the present examples are linear.

Nonlinear recursion cannot generally be reduced to

$$(\mathbb{1} - K)^{-1}.$$

Fourth, no fundamental constants, particle masses, or new experimental effects are derived here.

Fifth, the construction does not derive quantum mechanics.

Hilbert spaces, complex amplitudes, Green functions, and scattering operators enter the examples through their established physical theories.

The purpose of the architecture is to organize recursive structure **within** those theories.

33. Falsifiable Criteria for the PIHR Program

The proposed architecture should be judged by explicit criteria.

It would gain significance if:

1. physically different problems can be reduced by the **same factorization rules** without problem-specific modifications;
2. complicated path or diagram expansions can be compressed to substantially smaller kernels;
3. the minimal internal kernel can be constructed systematically;
4. nontrivial holonomy can be separated from ordinary dynamical propagation;
5. the resulting reduction yields analytic or computational advantages over standard formulations.

Conversely, the program would lose significance if:

1. every application requires an ad hoc redefinition of P, I, H, R ;

2. the kernel is no smaller or simpler than the original problem;
3. the factorization contains no information beyond standard transfer-matrix notation;
4. no new equivalence classes, invariants, algorithms, or deductions emerge from the framework.

These criteria make the proposal testable.

34. Conclusions

We introduced a four-component architecture for recursive Green-function and wave-scattering problems:

P = propagation,

I = interaction,

H = closed-path monodromy or holonomy,

and

R = recursive closure.

The first three components generate an irreducible return kernel

$$K_{PIH},$$

while the fourth performs closure:

$$\boxed{R(K_{PIH}) = (\mathbb{1} - K_{PIH})^{-1}.}$$

Under a unique factorization of admissible histories, we proved

$$\boxed{\mathcal{O} = \mathcal{O}_0 + F(\mathbb{1} - K_{PIH})^{-1}E.}$$

This produces an exact finite-kernel compression whenever the internal state space is finite:

$$\boxed{\text{infinitely many histories} \rightarrow \text{finite } K_{PIH} \rightarrow \text{one resolvent}.}$$

Seven standard physical constructions were then organized within the same framework.

Dyson dressing becomes

$$G = P R(IP).$$

Lippmann–Schwinger scattering becomes

$$T = I R(PI).$$

The Bethe–Salpeter scattering equation has the same recursive form.

Foldy–Lax multiple scattering becomes a finite matrix closure

$$\mathbf{u} = R(PI)\mathbf{u}^{(0)}.$$

Fabry–Pérot interference and quantum double-barrier tunneling reduce to the same scalar return kernel

$$r_1' r_2 e^{2ikL}.$$

An Aharonov–Bohm ring yields a matrix kernel in which the H sector contains genuine gauge holonomy.

The result therefore supports the structural correspondence

Dyson, Lippmann–Schwinger, Bethe–Salpeter, multiple scattering,
Fabry–Pérot, resonant tunneling, Aharonov–Bohm networks



$$(P, I, H) \rightarrow K_{PIH} \rightarrow R(K_{PIH}).$$

The individual formulas are established physics.

The proposed insight is that their repeated-history content belongs to a common architectural class.

The next decisive question is therefore no longer whether PIHR can reproduce these known examples.

It is whether the same calculus can solve a problem for which the conventional history expansion is sufficiently complicated that the architectural reduction produces a **genuine new analytic or computational shortcut**.

That constitutes the strongest test of the framework.

Declarations

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Conflict of Interest

The author declares no conflict of interest.

Data Availability

No datasets were generated or analyzed in this theoretical study.

Code Availability

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Appendix A. Nontrivial Compression Example

Exact PIHR Reduction of an Exponentially Proliferating Multi-Cycle Quantum Network

A.1. Purpose of the example

The resummation theorem developed in the main text establishes that recursively generated histories can be represented by a finite return kernel whenever the histories admit an appropriate factorization. The purpose of this appendix is to demonstrate that principle in a genuinely nontrivial system in which the number of individual paths grows exponentially.

We consider a three-arm quantum network of θ -graph topology. Two junctions, denoted L and R , are connected by three distinct internal arms. One external lead enters the graph at L , and a second external lead exits at R .

This system is particularly useful because it simultaneously contains:

- several independent propagation channels;
- nontrivial scattering and channel mixing at the junctions;
- two independent cycles and therefore nontrivial closed-path phase information;
- arbitrarily many internal round trips;
- and an exponentially increasing number of individual scattering histories.

Quantum graphs admit exact scattering and Green-function formulations in terms of propagation along bonds and local vertex-scattering amplitudes [19],[20],[21],[22],[24],[25]. Magnetic phases associated with closed paths can additionally be incorporated through the usual gauge-holonomy construction [29].

The central result below is that an infinite family of exponentially proliferating histories is represented exactly by a single 3×3 PIHR return kernel.

A.2. Three-arm θ -network

Let the three internal arms be indexed by

$$j = 1, 2, 3.$$

Their lengths are

$$L_1, \quad L_2, \quad L_3.$$

A wave propagating from the left junction to the right junction along arm j acquires a dynamical phase

$$e^{ikL_j}.$$

If the network is threaded by magnetic flux, define the directed gauge phase

$$\alpha_j = \frac{q}{\hbar} \int_L^R \mathbf{A} \cdot d\mathbf{l}$$

along arm j .

Forward propagation is therefore represented by

$$D_{\text{fwd}} = \text{diag}(e^{ikL_1+i\alpha_1}, e^{ikL_2+i\alpha_2}, e^{ikL_3+i\alpha_3}),$$

whereas reverse propagation is

$$D_{\text{rev}} = \text{diag}(e^{ikL_1-i\alpha_1}, e^{ikL_2-i\alpha_2}, e^{ikL_3-i\alpha_3}).$$

It is useful to separate propagation and holonomy explicitly. Define

$$P = \text{diag}(e^{ikL_1}, e^{ikL_2}, e^{ikL_3})$$

and

$$H_{\text{fwd}} = \text{diag}(e^{i\alpha_1}, e^{i\alpha_2}, e^{i\alpha_3}),$$

with

$$H_{\text{rev}} = H_{\text{fwd}}^{-1}.$$

Then

$$D_{\text{fwd}} = PH_{\text{fwd}},$$

and

$$D_{\text{rev}} = PH_{\text{rev}}.$$

Thus the dynamical propagation and gauge sectors remain explicitly separated.

A.3. Independent holonomies

The θ -graph has three arms but only two independent cycles.

A closed path travelling from L to R along arm i and returning along arm j acquires the gauge phase

$$e^{i(\alpha_i - \alpha_j)}.$$

Define

$$\phi_{12} = \alpha_1 - \alpha_2,$$

and

$$\phi_{13} = \alpha_1 - \alpha_3.$$

The third loop phase satisfies

$$\phi_{23} = \phi_{13} - \phi_{12}.$$

Hence only two holonomies are independent.

This is important for the PIHR interpretation. The H sector is not merely another local phase attached to propagation. It stores information about the two independent closed cycles of the network.

A common shift

$$\alpha_j \rightarrow \alpha_j + \chi$$

for every arm changes no closed-loop phase because

$$(\alpha_i + \chi) - (\alpha_j + \chi) = \alpha_i - \alpha_j.$$

The physically relevant information in H is therefore contained in phase differences associated with closed cycles.

A.4. Interaction sector

Let

$$S_L$$

be the internal scattering matrix at the left junction and

$$S_R$$

the corresponding matrix at the right junction.

Each is a 3×3 matrix acting on the internal arm amplitudes.

The matrix element

$$(S_R)_{ji}$$

is the amplitude for a wave arriving at R from arm i to be redirected into the reverse direction along arm j .

Likewise,

$$(S_L)_{ij}$$

maps a wave returning to L through arm j into a new forward propagation along arm i .

The interaction sector is therefore

$$I = \{S_L, S_R\}.$$

Let

$$a = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$$

denote injection from the left external lead into the three internal arms.

Let

$$b^T = (b_1 \quad b_2 \quad b_3)$$

describe extraction from the three arms into the right external lead.

A.5. One complete internal return

Consider a state immediately after it has left junction L toward R .

One complete internal return consists of the following sequence:

$$L \rightarrow R \rightarrow L.$$

In operator form, the sequence is:

32. forward propagation from L to R ;
33. internal scattering at R ;
34. backward propagation from R to L ;

35. internal scattering at L .

The resulting return operator is

$$\boxed{K_{PIH} = S_L D_{\text{rev}} S_R D_{\text{fwd}}.}$$

Separating propagation and holonomy gives

$$\boxed{K_{PIH} = S_L P H_{\text{rev}} S_R P H_{\text{fwd}}.}$$

This is the primitive PIH return kernel of the θ -network.

It contains all three primitive architectural sectors:

P :propagation along the three arms,

I :junction scattering,

H :closed-cycle gauge phase.

Importantly, K_{PIH} is only a

$$3 \times 3$$

matrix.

A.6. Paths with no complete internal return

A particle can enter the graph through L , propagate along one of the three arms, and leave through the right external lead without performing an additional round trip.

The corresponding transmission amplitude is

$$t^{(0)} = b^T D_{\text{fwd}} a.$$

In components,

$$t^{(0)} = \sum_{i_0=1}^3 b_{i_0} e^{ikL_{i_0} + i\alpha_{i_0}} a_{i_0}.$$

There are therefore

$$\boxed{N_0 = 3}$$

individual forward histories.

A.7. One complete internal round trip

A trajectory performing one complete return first travels along some forward arm i_0 .

At R , it is redirected into one of three backward arms j_1 .

At L , it is redirected again into one of three forward arms i_1 .

It then exits through R .

The total amplitude is

$$t^{(1)} = b^T D_{\text{fwd}} S_L D_{\text{rev}} S_R D_{\text{fwd}} a.$$

Using the return kernel,

$$\boxed{t^{(1)} = b^T D_{\text{fwd}} K_{PIH} a.}$$

In fully expanded form,

$$t^{(1)} = \sum_{i_0, j_1, i_1} b_{i_1} (D_{\text{fwd}})_{i_1 i_1} (S_L)_{i_1 j_1} (D_{\text{rev}})_{j_1 j_1} (S_R)_{j_1 i_0} (D_{\text{fwd}})_{i_0 i_0} a_{i_0}.$$

There are three independent choices for each of

$$i_0, \quad j_1, \quad i_1.$$

Therefore

$$\boxed{N_1 = 3^3 = 27.}$$

The matrix product K_{PIH} has already summed all 27 histories.

A.8. Two complete internal round trips

With two complete returns,

$$t^{(2)} = b^T D_{\text{fwd}} K_{PIH}^2 a.$$

An individual history is specified by

$$i_0, \quad j_1, \quad i_1, \quad j_2, \quad i_2.$$

Therefore

$$N_2 = 3^5.$$

Thus

$$\boxed{N_2 = 243.}$$

Again, all 243 individual trajectories are contained automatically in one matrix product:

$$K_{PIH}^2.$$

A.9. General path count

For n complete internal round trips, an individual history is characterized by

$$n + 1$$

forward-arm choices and

$$n$$

backward-arm choices.

Hence the number of histories is

$$N_n = 3^{n+1} 3^n.$$

Therefore

$$\boxed{N_n = 3^{2n+1} = 3 \cdot 9^n.}$$

The growth is exponential.

The cumulative number of histories containing at most n complete round trips is

$$N_{\leq n} = \sum_{m=0}^n 3 \cdot 9^m.$$

Summation gives

$$N_{\leq n} = \frac{3(9^{n+1} - 1)}{8}.$$

Several values illustrate the rapid proliferation:

Maximum round-trip number	Histories at that order	Cumulative histories
0	3	3
1	27	30
2	243	273
3	2,187	2,460
4	19,683	22,143
6	1,594,323	1,793,613
8	129,140,163	145,282,683

Thus, already through eight internal round trips, explicit enumeration requires more than
145 million

individual histories.

Allowing arbitrary round trips produces infinitely many.

A.10. PIHR compression

Despite this exponential proliferation, the amplitude containing exactly n returns is simply

$$t^{(n)} = b^T D_{\text{fwd}} K_{PIH}^n a.$$

The complete transmission amplitude is therefore

$$t = \sum_{n=0}^{\infty} t^{(n)}.$$

Hence

$$t = b^T D_{\text{fwd}} \left[\sum_{n=0}^{\infty} K_{PIH}^n \right] a.$$

Applying recursive closure,

$$R(K_{PIH}) = \sum_{n=0}^{\infty} K_{PIH}^n.$$

Therefore

$$t = b^T D_{\text{fwd}} R(K_{PIH}) a.$$

Whenever the Neumann series converges,

$$R(K_{PIH}) = (\mathbb{1} - K_{PIH})^{-1}.$$

Consequently,

$$t = b^T D_{\text{fwd}} (\mathbb{1} - K_{PIH})^{-1} a.$$

This is the exact all-orders result.

The complete infinite path family has therefore been compressed into the inverse of a

$$3 \times 3$$

matrix.

A.11. Nontrivial Compression Proposition

Proposition A1 — Exponential-to-Finite PIHR Compression

Consider a three-arm Θ -network with nonzero internal transition amplitudes between every pair of arms at both junctions. Let K_{PIH} be the three-dimensional return kernel

$$K_{PIH} = S_L P H_{\text{rev}} S_R P H_{\text{fwd}}.$$

Then the number of distinct histories containing exactly n complete internal returns is

$$N_n = 3 \cdot 9^n,$$

whereas their complete coherent amplitude is represented by

$$b^T D_{\text{fwd}} K_{PIH}^n a.$$

Summing over all recursive depths gives

$$t = b^T D_{\text{fwd}} (\mathbb{1} - K_{PIH})^{-1} a.$$

Therefore

$$N_n \xrightarrow{n \rightarrow \infty} \infty$$

while

$$\dim K_{PIH} = 3$$

remains constant.

Proof. For every complete return, scattering at R allows three possible backward arms and scattering at L allows three subsequent forward arms. Thus each additional return multiplies the number of histories by

$$3 \times 3 = 9.$$

The initial propagation allows three choices, giving

$$N_n = 3 \cdot 9^n.$$

Matrix multiplication distributes automatically over all intermediate channel indices.

Consequently,

$$K_{PIH}^n$$

contains the coherent sum over every sequence of n internal return blocks.

Summing all recursive depths yields the Neumann series and hence the resolvent. ■

A.12. General m -arm result

The same argument extends immediately to an m -arm two-junction network.

The internal PIHR state space has dimension

$$m.$$

For complete junction mixing, the number of histories containing n complete internal returns is

$$N_n = m^{2n+1}.$$

Nevertheless,

$$K_{PIH} \in M_m(\mathbb{C}),$$

and

$$t = b^T D_{\text{fwd}} (\mathbb{1} - K_{PIH})^{-1} a.$$

Hence

$$N_n \sim m^{2n}$$

grows exponentially, whereas the kernel dimension remains

$$d = m.$$

This provides a general exact compression relation:

$$\text{exponential path proliferation} \rightarrow \text{fixed-dimensional recursive representation.}$$

A.13. Explicit Kirchhoff-junction realization

A particularly transparent realization is obtained using identical Kirchhoff-type junctions.

Each junction has four attached channels:

- one external lead;
- three internal arms.

For a degree-four Kirchhoff junction, the vertex-scattering matrix has the familiar form

$$\sigma = \frac{2}{4} J_4 - I_4.$$

The internal 3×3 block is therefore

$$S = \frac{1}{2} J_3 - I_3.$$

Explicitly,

$$S = \frac{1}{2} \begin{pmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{pmatrix}.$$

Injection from the left lead is described by

$$a = \frac{1}{2} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix},$$

and extraction into the right lead by

$$b^T = \frac{1}{2} (1 \quad 1 \quad 1).$$

For identical junctions,

$$S_L = S_R = S.$$

A.14. Equal-length network

Set

$$L_1 = L_2 = L_3 = L.$$

Choose a gauge in which

$$H = \text{diag}(1, e^{i\phi_1}, e^{i\phi_2}).$$

Then

$$D_{\text{fwd}} = e^{ikL} H,$$

and

$$D_{\text{rev}} = e^{ikL} H^{-1}.$$

The PIHR return kernel becomes

$$\boxed{K_{PIH} = e^{2ikL} S H^{-1} S H.}$$

The exact transmission amplitude is

$$\boxed{t(k, \phi_1, \phi_2) = \frac{e^{ikL}}{4} \mathbf{1}^T H [\mathbb{1} - e^{2ikL} S H^{-1} S H]^{-1} \mathbf{1},}$$

where

$$\mathbf{1} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

Thus the exact transmission through a graph containing two independent cycles and infinitely many possible windings requires only one 3×3 inverse.

A.15. Gauge invariance

Consider the common transformation

$$H \rightarrow H' = e^{i\chi} H.$$

Then

$$(H')^{-1} = e^{-i\chi} H^{-1}.$$

The return kernel transforms as

$$K'_{PIH} = e^{2ikL} S (H')^{-1} S H'.$$

Therefore

$$K'_{PIH} = e^{2ikL} S e^{-i\chi} H^{-1} S e^{i\chi} H.$$

The scalar phases cancel:

$$\boxed{K'_{PIH} = K_{PIH}.}$$

The transmission amplitude acquires at most the irrelevant overall phase

$$t' = e^{i\chi} t,$$

so that

$$\boxed{|t'|^2 = |t|^2.}$$

Thus the PIHR representation automatically retains only the physically relevant relative closed-cycle phases.

A.16. Resonances

The recursive amplification becomes singular when

$$\mathbb{1} - K_{PIH}$$

fails to be invertible.

The resonance condition is therefore

$$\boxed{\det(\mathbb{1} - K_{PIH}) = 0.}$$

For the equal-length Kirchhoff network,

$$\boxed{\det[\mathbb{1} - e^{2ikL} S H^{-1} S H] = 0.}$$

Thus the same 3×3 kernel simultaneously determines:

- the complete multiple-path transmission amplitude;
- the interference between different cycles;
- the dependence on the two independent holonomies;
- and the resonance condition.

No individual path enumeration is required.

For real-energy scattering, the retarded prescription may be implemented by taking

$$k \rightarrow k + i\varepsilon, \quad \varepsilon \downarrow 0.$$

Equivalently, the finite-dimensional resolvent may be regarded as the meromorphic continuation of the convergent expression away from its poles.

A.17. Why this example is genuinely nontrivial

The θ -network differs qualitatively from the scalar Fabry–Pérot and double-barrier examples.

In those systems the return kernel is a single complex number.

Here,

$$K_{PIH}$$

is a matrix.

Furthermore,

$$S$$

and

$$H$$

do not generally commute:

$$SH \neq HS.$$

Consequently, the result cannot in general be reduced to a scalar geometric series before the internal channel structure is solved.

All four PIHR components are necessary:

P :three-channel propagation,

I :nontrivial junction mixing,

H :two independent closed-cycle phases,

R :closure over arbitrarily many internal windings.

This is therefore a genuinely matrix-valued PIHR problem.

A.18. Quantitative compression

The distinction between explicit path enumeration and PIHR reduction can now be stated quantitatively.

At order n ,

$$N_n = 3 \cdot 9^n.$$

For example,

$$N_8 = 129,140,163.$$

Through order eight,

$$N_{\leq 8} = 145,282,683.$$

Yet the exact solution including not merely eight but **all** round trips is represented by

$$(\mathbb{1} - K_{PIH})^{-1}, \quad K_{PIH} \in M_3(\mathbb{C}).$$

The compression can therefore be summarized as

145,282,683 histories through eight returns

$$\Downarrow$$

3×3 PIHR kernel

while the same matrix inverse simultaneously includes the histories of orders

$$9, 10, 11, \dots, \infty.$$

The relevant comparison is therefore stronger than a finite compression ratio.

It is

$$\boxed{\text{infinitely many histories} \rightarrow \text{nine kernel entries.}}$$

A.19. Architectural interpretation

The calculation illustrates the distinction between microscopic history complexity and architectural complexity.

Microscopic history complexity increases without bound:

$$\lim_{n \rightarrow \infty} N_n = \infty.$$

Architectural state complexity remains fixed:

$$\dim \mathcal{X} = 3.$$

Thus

$$\boxed{\text{history complexity} \neq \text{architectural complexity.}}$$

The PIHR kernel does not approximate or discard the individual histories.

Instead, matrix multiplication performs their coherent summation implicitly.

Recursive closure then performs the infinite resummation.

The reduction therefore proceeds as

$$\begin{aligned} & \boxed{\text{individual scattering histories}} \\ & \Downarrow \\ & \boxed{(P, I, H)} \\ & \Downarrow \\ & \boxed{K_{PIH} = S_L P H_{\text{rev}} S_R P H_{\text{fwd}}} \\ & \Downarrow R \\ & \boxed{(\mathbb{1} - K_{PIH})^{-1}.} \end{aligned}$$

A.20. Relation to the PIHR Resummation Theorem

This example supplies a direct nontrivial realization of Theorem 1 of the main manuscript.

The primitive return family consists of all elementary round trips

$$i \rightarrow j \rightarrow k,$$

where

$$i, j, k \in \{1, 2, 3\}.$$

There are

such elementary histories when entrance and subsequent forward propagation are explicitly distinguished.

Their amplitudes are not stored independently.

They are contracted into the matrix product

$$S_L D_{\text{rev}} S_R D_{\text{fwd}}.$$

Repeated composition generates

$$K_{PIH}^n.$$

Recursive closure then gives

$$R(K_{PIH}) = (\mathbb{1} - K_{PIH})^{-1}.$$

Accordingly, the example realizes the complete chain

unique history factorization $\rightarrow$ finite PIH return kernel $\rightarrow$ recursive closure $\rightarrow$ exact all-orders amplitude.

A.21. Significance for the proposed compression principle

The example allows the compression claim of the main manuscript to be stated in a quantitatively testable form.

The number of explicit trajectories satisfies

$$N_n = 3 \cdot 9^n,$$

while

$$\dim K_{PIH} = 3.$$

Therefore

$$\frac{N_n}{\dim K_{PIH}} = 9^n.$$

The ratio diverges exponentially:

$$\lim_{n \rightarrow \infty} \frac{N_n}{\dim K_{PIH}} = \infty.$$

Nevertheless, the exact infinite-history result remains finite-dimensional.

This establishes, for the present class of networks, the quantitative form of the proposed PIHR compression principle:

exponentially proliferating histories $\Rightarrow$ increasing representation dimension.

Instead,

recursive physical complexity $\rightarrow$ fixed-dimensional architectural closure.

A.22. Conclusion

The three-arm θ -network provides a nontrivial demonstration of the PIHR framework in which all four architectural components are simultaneously active.

Propagation occurs through three distinct channels.

Interaction mixes these channels at two junctions.

Holonomy records two independent closed-cycle phases.

Recursive closure sums arbitrarily many internal returns.

The number of explicit scattering histories grows as

$$3 \cdot 9^n,$$

but the exact all-orders transmission amplitude is

$$t = b^T D_{\text{fwd}} (\mathbb{1} - K_{PIH})^{-1} a$$

with a fixed

$$3 \times 3$$

return kernel.

Thus the architecture performs the exact reduction

$$\text{infinitely many paths} \rightarrow \text{finite PIH kernel} \rightarrow \text{one matrix inverse.}$$

This example strengthens the interpretation of PIHR as more than a classification language. Within this class of recursive network problems, it functions as an exact compression calculus that separates local propagation, interaction, global closed-cycle information, and recursive history generation while retaining the complete coherent amplitude.